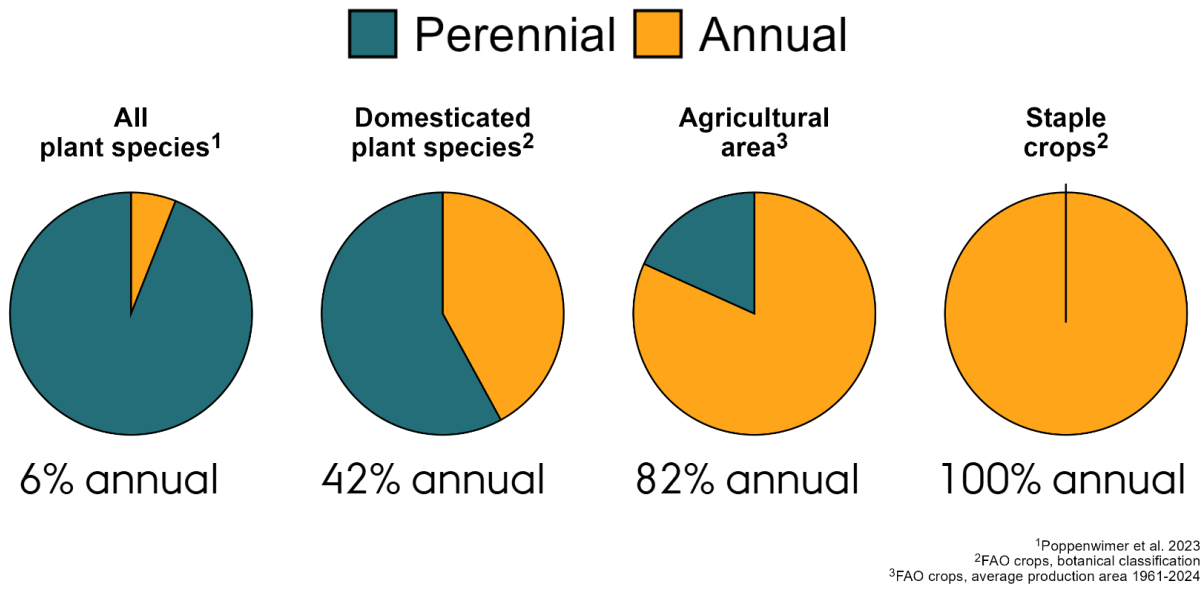
GoodGrain Home Baking Evaluation

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For the curious

Broadly speaking, all plants fall *roughly* into two categories: perennial and annual. Perennial plants live for more than one year, while annual plants do not. In nature, perennial plants are very common. Think of a neighborhood park, or the forest: there will be trees, shrubs, maybe some grass. These are all perennial plants. In contrast, many of the plants in your kitchen garden are annuals, meaning you have to re-plant them every year (for example radishes, lettuce, or cucumbers). This general pattern of '**perennials in nature, annuals in agriculture**' is actually seen at a global scale. And while we do have a lot of wonderful perennial crops (for example apples, avocados, and coffee), the crops we use the most (and the ones we plant on the most land) are annuals, as you can see in the figure below.

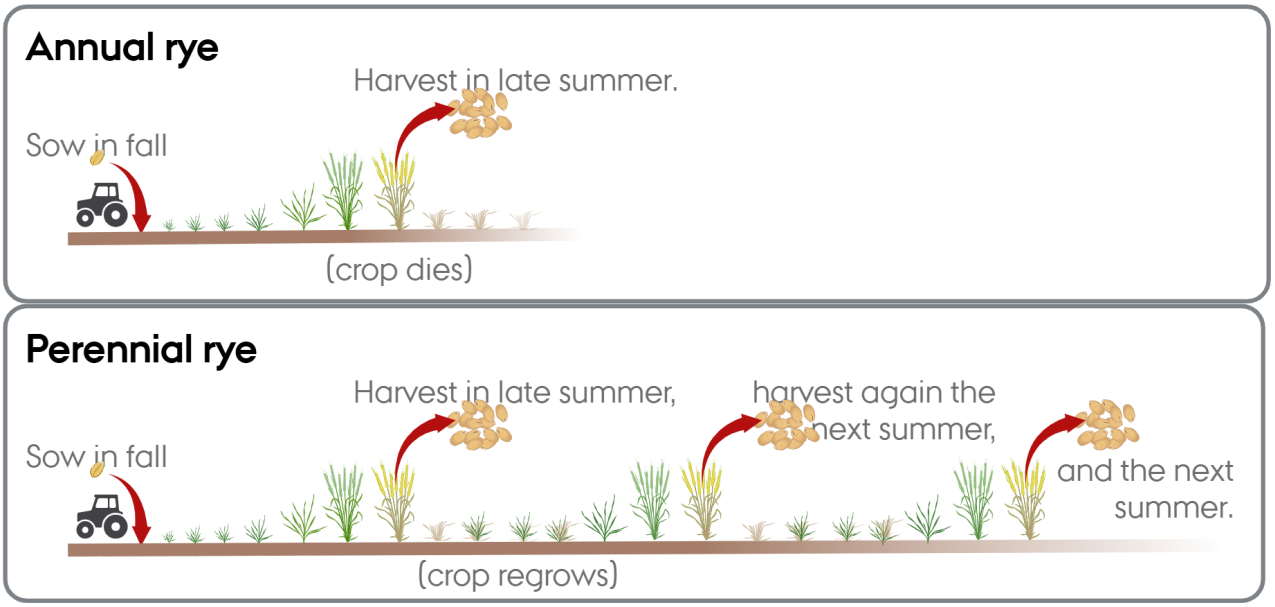


In nature only 6% of the plant species are annual. Early human farmers seemed to prefer annuals, as 42% of the plants we have domesticated in some way are annuals. While less than half of the plants we have domesticated are annuals, most of our agricultural land is planted to annuals. This is because most of the calories humans consume derive from annual crops such as maize, rice, potatoes, and wheat. Crops we commonly use to make breads in Denmark (wheat, oats, rye) are all annual crops. Figure created by Virginia Anne Nichols.

Are annual plants superior to perennials for food production? Did humans *deliberately* choose to domesticate annual crops? Or was it just coincidence? These questions are currently being debated. However, regardless of how we got here, it is clear that having more perennials integrated into our agricultural land would be good for the environment.

This is because **perennials tend to offer benefits to the overall environment**; their extensive roots allow them to use resources like water and nitrogen more efficiently, they help put carbon into the soil, and they provide soil cover year-round. Creating perennial versions of our staple crops, like the perennial rye used in this project, is one way to get more perennials onto our agricultural land.

Societies have long recognized the potential benefits of a perennial grain. **There have been efforts to breed perennial wheat in the Soviet Union since the 1930s**! In the past, it has been a very difficult task, mainly because scientists had few tools to see what was going on inside the plant's genetics. Despite these challenges, a few laboratories got lucky and had some successes, including one in Germany at the University of Hohenheim. There, in the 1960s, Dr. Reimann-Philips and his team successfully crossed an annual rye with a old perennial rye ancestor. The team tested the results, and made more selections. The resulting crop was perennial (see figure below), and in the 1990s it was submitted to be considered as a variety in Germany.



Annual rye is sown in the fall, and the crop dies in late summer before the grain is harvested. Perennial rye is also sown in the fall, and while the growth from that year dies and grain is harvested just like in the annual crop, the roots stay alive and the crop regrows after harvest. This new growth acts just like a crop that is sown in the fall…except you didn't have to sow it. Image made by Virginia Anne Nichols using Biorender.

The German government rejected the application for complicated reasons, and it was never released as a variety. The breeders maintained some plots as their passion project, and while some of the material was later released as a wildlife forage crop, the perennial rye hasn't received full attention for the past thirty years. In 2023, Dr. Thomas Miedaner, who took over the University of Hohenheim's rye breeding program after Dr. Reimann-Philips' passing, shared some seeds of this crop with several researchers at Aarhus University. We think the idea of a perennial rye has a lot of potential in Denmark!

In 2025 we tested the perennial rye side-by-side against a high-yielding annual rye. We were very impressed when it produced half of the grain yield of the modern annual rye, given that it was crossed with varieties from the 1960s! The perennial rye produced grain again in 2026, but we need more research to determine the optimal way to manage the crop. We think this variety deserves more development, and this grant is part of that effort. One of the goals of the grant is to understand how this new perennial rye performs in home kitchens, and to understand how people feel about eating a perennial grain.

Funding and project partners

This project is part of a **Novo Nordisk Fonden Plant2Food** research project (Grant #NNF22SA0081019) being conducted at the Department of Agroecology and Department of Food Science at **Aarhus University** in Denmark in collaboration Wageningen University and Research in the Netherlands, and with Meyers, a prominent Danish food enterprise. As mentioned above, you can read a bit more about the GoodGrain project here.